

Highlights

Combating Digital Media Piracy with Agentic AI: Leveraging Video Transcription and Character Recognition for Automated Enforcement

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- The proposed AI Agent framework demonstrates high scalability, enabling dynamic adaptation to evolving piracy tactics and varying workload demands.
- Automated transcription matching capture at pre-determined time interval and image matching with original content transcription strengthen verification, significantly reducing human intervention in the anti-piracy workflow and potential piracy control for dubbed videos.
- Automated snapshot capture at pre-determined time interval and image matching with original content strengthen verification, significantly reducing human intervention in the anti-piracy workflow.
- Object Detection and mapping with original content object. A novel method because all reviewer might not have been trained on the video content but will definitely know the characters associated.
- The system autonomously scrapes and accurately parses relevant URLs from platforms such as YouTube as well as from websites containing both static and dynamic content, thus broadening its coverage of potential infringement sources with more than 90% accuracy.
- This technology empowers media companies to combat piracy more effectively, unlocking substantial additional revenue streams by minimizing unauthorized content distribution.
- By preserving intellectual property and supporting legitimate content distribution, the framework contributes to increased market stability and sustainable economic growth within the digital media sector.

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Abstract

Media piracy—the unauthorized reproduction and distribution of digital content—remains a threat to the media industry’s. This pervasive issue results in substantial annual revenue losses, estimated between *29billionand71* billion, thereby hampering economic growth and diminishing incentives for creative innovation. Implementing robust anti-piracy measures not only safeguards revenue streams but also enhances consumer migration to legitimate Over-the-Top (OTT) platforms such as Netflix and Zee, fostering a healthier media ecosystem. This research paper introduces a novel Agentic Artificial Intelligence (AI) framework leveraging large language models (LLMs) for advanced piracy detection, particularly focusing on video transcription and character matching. Unlike previous approaches that mainly analyze video frames, the proposed methodology incorporates transcription-based matching to effectively differentiate full-length pirated content from trailers or excerpts. This is particularly advantageous in detecting dubbed versions, where conventional frame-matching techniques often fall short. Video character matching further augments content verification, especially for reviewers lacking familiarity with global media releases. The framework autonomously conducts large-scale analysis of both open and illicit platforms, employing automated copyright URL identification, dynamic web scraping (including sites with continuously updating HTML), platform login management, video playback assessment, and image-embedded object recognition. Heuristics are implemented to distinguish trailers from complete works, enhancing detection specificity. Experimental evaluations affirm the proposed system’s scalability, reliability, and marked reduction in required manual oversight. This Agentic AI approach, integrating LLM-driven video transcription analysis,

represents a significant advancement in the global fight against digital media piracy, providing an effective and adaptable solution for content protection.

Keywords: AI Agent, Anti-Piracy, Artificial Intelligence, Content Protection, Economic Benefits of Antipiracy, Function calling, GenAI, Large Language Model, Open AI GPT-4 , Video Audio Transcription, google gemini flash

1. INTRODUCTION

Fig. 2 shows the proposed framework for anti-piracy using agentic AI. This research searches for piracy in YouTube video uploads. Media piracy refers to the unauthorized copying, distribution, and consumption of digital content, such as movies, music, television shows, and e-books, without permission from the copyright holders. This illicit activity has widespread impacts on society and the media industry, including significant revenue loss of almost \$29 billion and \$71 billion annually as mentioned in [1]. As piracy grows more sophisticated, technological intervention becomes urgently necessary. Automation and advanced tools are crucial for detecting, preventing, and rapidly responding to infringing activities across multiple platforms and geographies.

Organizations have relied on full video matches , content id match, watermarks as put forward in [2]. While these approaches offer some mitigation, they often lack adaptability, scalability, and the capability to understand evolving piracy tactics. Current youtube copyright detection tool [2] does not talk about using video transcription as a matching technique considering altering videos in different languages captions different languages as in Fig. 1.

This study proposes an Anti-Piracy AI Agent that is scalable and can accurately scrape websites in a very short time, and also take snapshots, celebrity recognition through LLM, Video transcription for checking dubbed content and later compare them to detect the sanity, which leads to very little human intervention. In this research, we have used Langchain AI Agent with Open AI ChatGPT for Agent, video translation, Gemini [3] for video personality recognition and a storage for snapshots. This research also simulates two dynamic websites that change their HTML to avoid getting detected, thus making it difficult for traditional rule-based web scraping systems to produce better results. It also enhances easy report format changes, thus making it

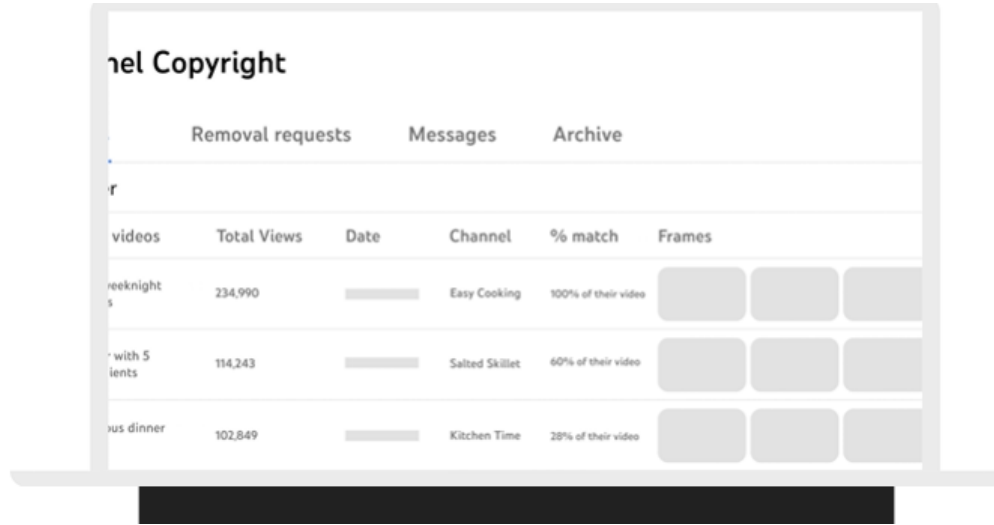


Figure 1: YouTube Copyright Tool Snapshot

a global product. This research illustrates and proves that a vast amount of the reasoning and actions required for detecting piracy can be achieved through AI Agents integrated with different tools and large language models, as shown in Fig. 2

1.1. BACKGROUND OF STUDY

Organizations have relied on full video matches , content id match, watermarks as put forward in [2].The new AI Anti-Piracy agent will be able to extract from dynamic websites, making it more flexible. In terms of humans verifying the images, the AI Agent can compare the images and detect similar objects and mark the content as pirated.The AI Agent will also do a transcription of the video and match the content. This is a very useful tool in case of videos being tranformed to other languages which makes this research a novel.

1.2. PROBLEM STATEMENT

Conventional media anti-piracy approaches are siloed, separate, non-adaptive scripts or services that handle each task (scraping, taking snapshots, report creation, action-taking). This results in slow adaptation to new piracy techniques, increased operational overhead, and limited end-to-end automation.

Figure 2: Anti-Piracy Agentic AI Framework

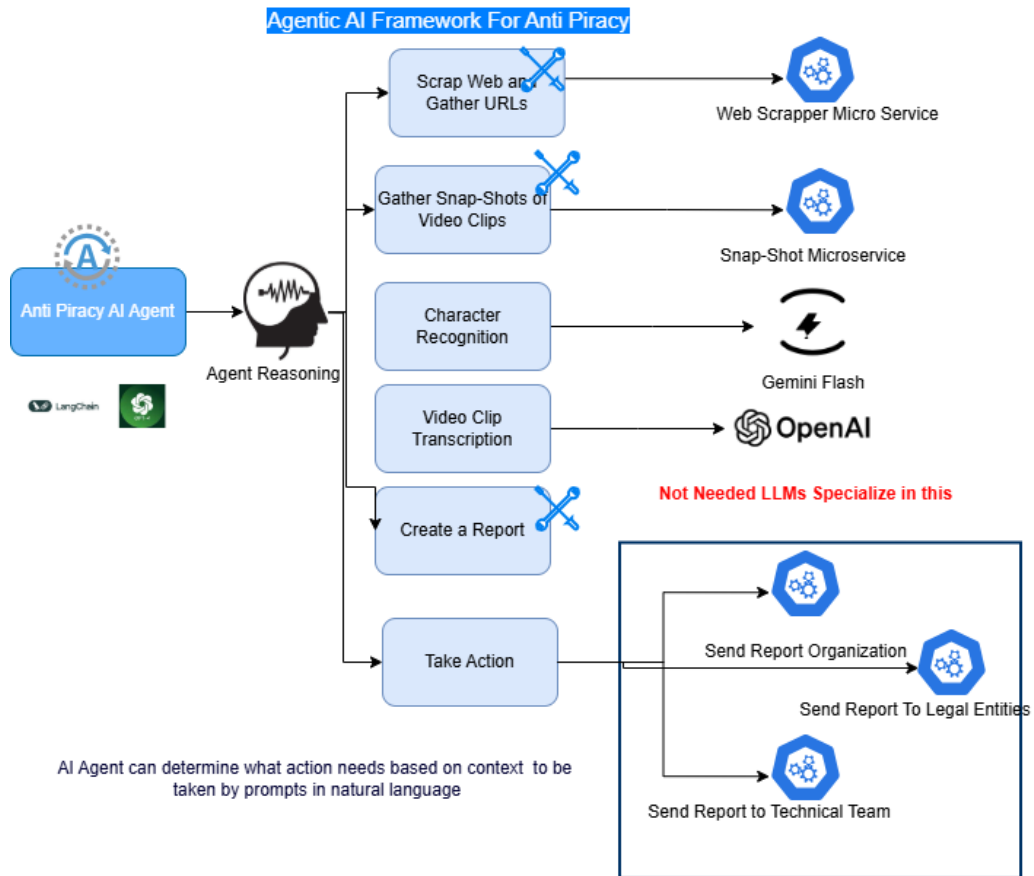


Figure 3: Anti-Piracy Agentic AI Framework

Feature	Traditional Scraper	AI Scraper Agent (LangChain LLM)
HTML structure dependent	✓ Yes	✗ No – adapts based on semantics
Handles JS-rendered content	✗ No (needs workaround)	✓ Yes (via Playwright tools)
Adaptive retry logic	✗ Manual	✓ Built-in via reasoning
Works with login/session	⚠ Requires setup	✓ Easy with planning tools
Understands page content	✗ No	✓ Yes (via language understanding)

Figure 4: Traditional Vs AI-Based Piracy Agent Comparison

1.3. Research Questions

1. How does an AI agent outperform traditional Python/microservice-based anti-piracy solutions in efficiency and adaptability?
2. Can AI agents autonomously determine and execute contextually appropriate actions in a dynamic piracy landscape?
3. What roles do LLMs and AI agents play in automating report generation and decision-making processes?
4. How many urls can the Ai Agent correctly report and what is the time taken
5. How can we scale the agent?
6. Can the AI Agent successfult transcribe and summarize the audio in the video?

2. RELATED WORK

Recent works like [4] highlight the rise of agentic AI frameworks (e.g., LangChain agents) that integrate LLM-based reasoning with traditional software components, enabling cognitive workflows. Conventional microservices alone are shown to be rigid and require frequent rewiring for new threats. [5] have brought forward some research on blocking domain name addresses.[6] discuss anti-piracy and its potential regional and social biases. [7] discuss

using a third-party integration to existing traditional scripts . In contrast, LLM-driven agents are being explored in cybersecurity and compliance for their ability to synthesize information, draw actionable insights, and can call required Langchain tools which can internally take actions by calling required APIs to complete the task. The depicted framework exemplifies this shift: LLM-powered reasoning orchestrates specialized services, streamlining anti-piracy operations and reducing development burden.

2.1. LITERATURE REVIEW GAPS

None of the reports focuses on transcription and personality recognition of pirated video using AI Agents and Large Language Models in Anti-Piracy for media.

3. RESEARCH METHODOLOGY

3.1. System Design and Integration

Architecture: Create a Langchain Agent using OpenAI GPT-4 and related tools to detect and report piracy. This research searches for piracy in YouTube video uploads. Though YouTube has strict regulations against piracy control, this will be just one of the examples taken to demonstrate the proof of concept. For this purpose, we have taken the famous Hollywood movie, Titanic, to search for piracy on YouTube. This research also tested and simulated websites with dynamic HTML using React as a frontend application.

3.2. Data Collection

Data Used: This Piracy Agent accepts Movie Names and Series Names. For this purpose, we have taken the example of the famous Hollywood movie Titanic, which was used to search for piracy on YouTube. The tool can be generalized for websites with dynamic content changes.

3.3. Evaluation

The evaluation of this study will be as follows, considering five movie names and two simulated pirated dynamic websites.

- Number of infringing URLs detected
- Number of accurate snapshot verifications

- Reduction in human intervention (if measured) by generating transcriptions of video clips which can then be matched
- Personality recognitions from Movie Clips

4. EXPERIMENTAL SETUP AND PROPOSED METHOD

4.1. Code Availability And Description

The source code and output for the experiments in this paper are available at [8][9]. The technology used is listed in Table 1. This Anti-Piracy Agent is built with GTP-4 and a set of tools, each with its own function and prompts to take specific actions, scraping the web, taking snapshots, and creating reports. One of the tool used for personality detection in video is using Gemini[3].

5. RESULTS

Table 1: Video Dataset Summary 17/06/2025

Title	Duration	Publish Time	URL	Character	Transcription	Notes
Kabhi Khushi Kabhi Gam	03:29:51	5 years ago	https://www.youtube.com/watch?v=9iH1BCD8ORc	Shahrukh Khan	Due to large content, the result can be viewed in the code repo	Movie Characters identified correctly
Titanic	0.29	year 2022	https://www.youtube.com/watch?v=ldMsNzPR6yY	–	Due to large content, the result can be viewed in the code repo	Trailer as shown from duration
Kungfu Panda	27:41:00	year 2025	https://www.youtube.com/watch?v=DvfTJqcZEgM	Po, the protagonist of the <i>Kung Fu Panda</i> franchise. He is voiced by Jack Black	Due to large content, the result can be viewed in the code repo	Identified the movie character
Simulated Dynamic Web 1	–	–	3 URLs	–	–	–
Simulated Dynamic Web 2	–	–	2 URLs	–	–	–

Below are few prompts given to the Agent for testing Given a user prompt to scrape YouTube as below

5.1. User Prompt 1

Get piracy report for Bollywood movie Kabhi Khushi kabhi Gam: First scrap and then take snapshots and then create report and get transcription. In the agent response include the movie name and scrapped url , duration of movie and name the character in focus in the movie. Eg, output:


```
{"movie name":"Dhoom1","movie link":"https://youtube...","characters":["Abhishek"]}
```

Response The Anti-Piracy Agent considers the next step to accomplish its task. For the above example, it reasons to follow the sequence of tool calls: `scrape_movie_links_tool` with the entity as movie name passed as tool input, `play_and_snapshot_tool` with the URL parsed in the previous step, then `capture_characters_faces_tool` which identifies personalities in the video clip, `transcription_summarization_tool` which creates the transcription of the video and `create_report_piracy` with the URL passed, which also ingests the snapshot folder and generates a report.

Author Observation As shown in Fig. 5 The Anti-Piracy AI Agent successfully took the snapshots. The agent also successfully identifies the personalities using Gemini Large Language Model. Upon which the agent also transcribed a clip successfully showing the ability of the agent to support detection of dubbed videos. Results are shown in Table.

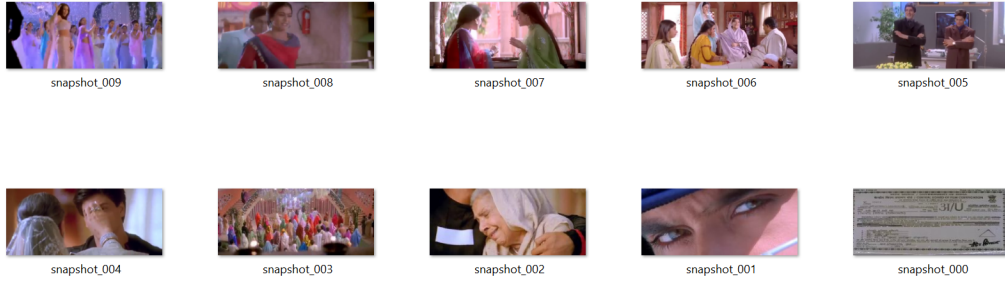


Figure 5: Snapshots from the movie Titanic captured by the Anti-Piracy AI Agent.

5.2. User Prompt2

Get all the links and generate report from the pirated websites with username as username, password as password and url as "http://localhost:5174/" and "http://localhost:5173/".

Author Observation The Anti-Piracy Agent successfully provided the required details for scraping the website and was able to navigate to the home page to extract the pirated links, as marked in the image. Code for the same can be found in [8]. Since we used simulated dummy websites with dynamic HTML changes, this also demonstrates that the agent could parse both sites

with little to no code changes. Regarding scaling, we can dockerize this container and scale it using AWS Auto Scaling Services, with a queue message count that includes movie names.

This study shows how an Anti-Piracy AI Agent can intuitively think of the following steps to detect piracy as shown in TABLE 1. It also shows how dynamically any number of movie names can be passed and scraped. The agent can find the next feasible step to the final answer. The agent also generated a response with URLs and snapshot image locations. This research can be extended to include more tools and steps for LLM for web series, which might need a different route to parse and detect piracy. If enabled, this agent can also be triggered to start the tasks on a natural language command with a list of movies or with voice. The Agent could also successfully scrape dynamic websites and fetch the URLs. Regular scraping and reporting can increase the review substantially for media companies. The results have been verified with the snapshots generated for pirated links and the scrapped urls in agent output.

6. Conclusion

As shown in TABLE 1, The AI Agent could successfully scrap movie links on YouTube, Get snapshots, detect personalities, transcribe the video to detect dubbed content and also generate reports and also reason out the next step which it needs to take. Combining the all the mentioned steps this agent is novel anti-piracy agent.

7. Funding

No funds were received to carry out this research.

8. Funding

No funds were received to carry out this research.

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